A Plan for Community Sovereign Kiosks on Minjerribah

Enhancing Island Resilience and Self-Reliance



A Proposal for the Straddie Chamber of Commerce.
Please note: This is a plan only; nothing is registered.

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Understanding Minjerribah's Resilience Challenges

North Stradbroke Island (Minjerribah) faces a unique set of escalating risks that could lead to periods of isolation from the mainland.



Climate Risks

Increased frequency and intensity of cyclones and bushfires.



Systemic Risks

Potential for civilisation-level disruptions, such as major solar flares or coordinated cyber attacks.



Infrastructure Vulnerabilities

In such scenarios, mainland links for power, communications, food supply, and transport (ferries) could be severed for extended periods.

The island's planning considers the possibility of operating under total isolation and grid-down conditions.

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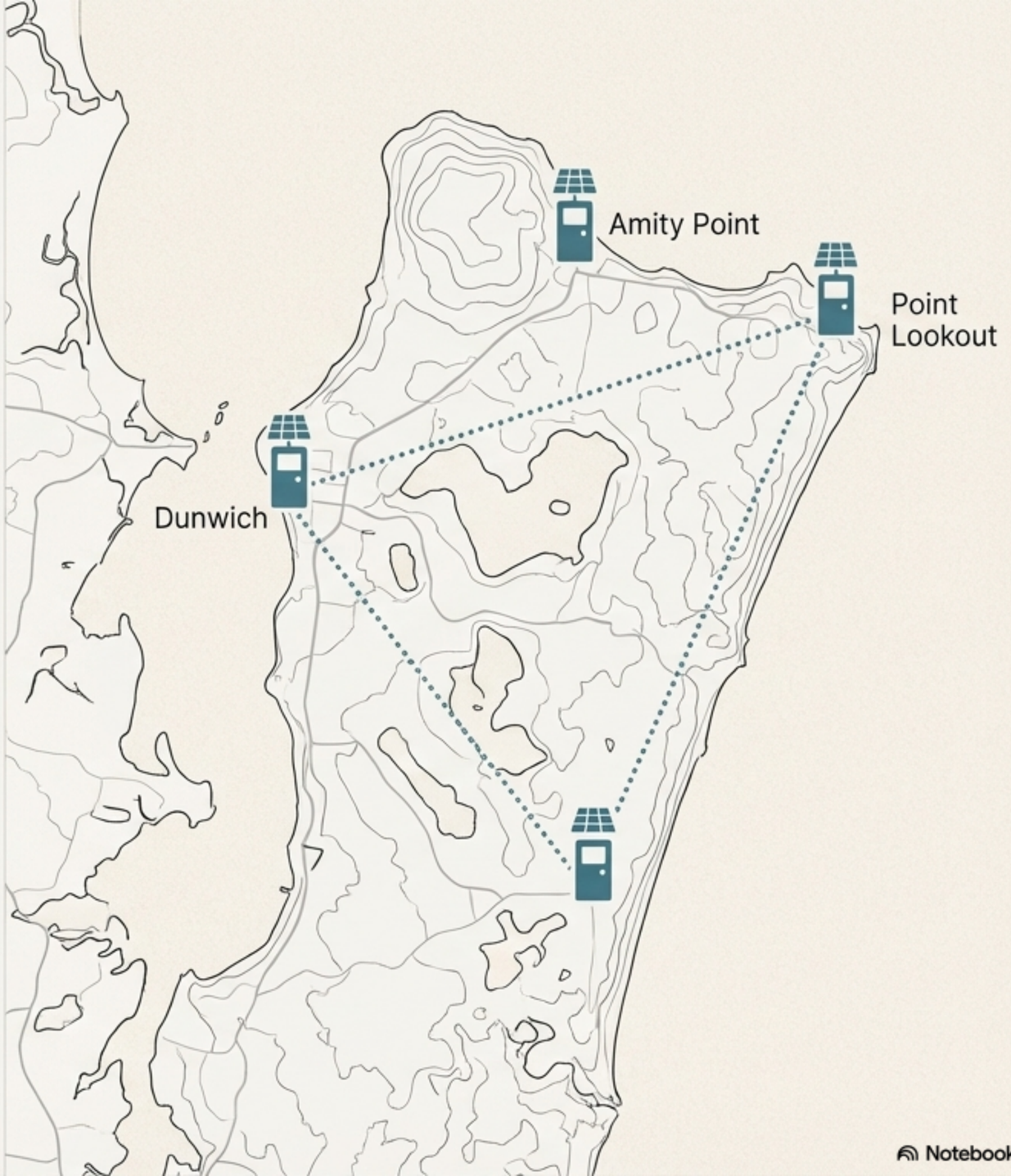
The Proposal: A Network of Community Sovereign Kiosks

Core Concept

This plan proposes a network of offline-first, solar-powered community kiosks, each functioning as a “Sovereign Node.”

Explanation

- These are independent, resilient hubs designed to keep critical services running locally, even when all outside systems fail.
- Each kiosk is a compact unit (about the size of an ATM) that operates without needing external power or an internet connection.
- The network is designed to enhance community self-reliance and support Quandamooka data sovereignty.



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Kiosk Design: Engineered for Island Conditions

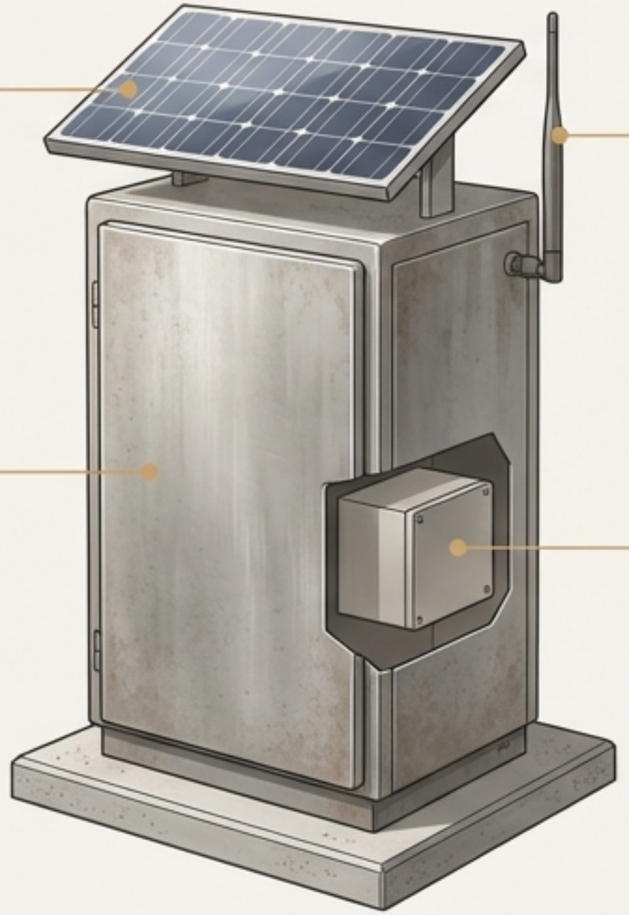
Each kiosk is a standalone, micro off-grid facility built to infrastructure-grade standards. Community input and Indigenous knowledge of local conditions will guide final site selection.

1 Independent Power

Solar PV panels and deep-cycle battery storage, sufficient for several days of operation without sun.

2 Hardened Enclosure

Steel or composite housing, anchored above flood lines to withstand Category D cyclone winds and bushfire heat.



3 Rugged Communications

Built-in long-range radio (e.g., LoRa mesh, UHF/VHF) and reinforced antennas for kiosk-to-kiosk communication when other networks are down.

4 Resilient Electronics

Low-power, efficient processors with critical components housed in Faraday-shielded casings to improve survivability against electromagnetic pulses (EMP).

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The Technology: How the Kiosks Work Offline

Core Principle

The network uses an offline-first, peer-to-peer software architecture. Kiosks function autonomously and synchronise data with each other whenever a connection is possible.

How it Works

- **Conflict-Free Data Syncing (CRDTs):** A method for kiosks to share and merge information after being offline. This ensures all data (like messages or votes) is consistent across the network without conflicts.
- **Peer-to-Peer Networking:** Kiosks communicate directly using local radio or Wi-Fi. If they can't connect wirelessly, data can be physically transferred between them (a 'sneakernet').
- **Secure & Decentralised Identity (DIDs):** Users have cryptographic digital IDs to sign information, ensuring authenticity without a central authority.

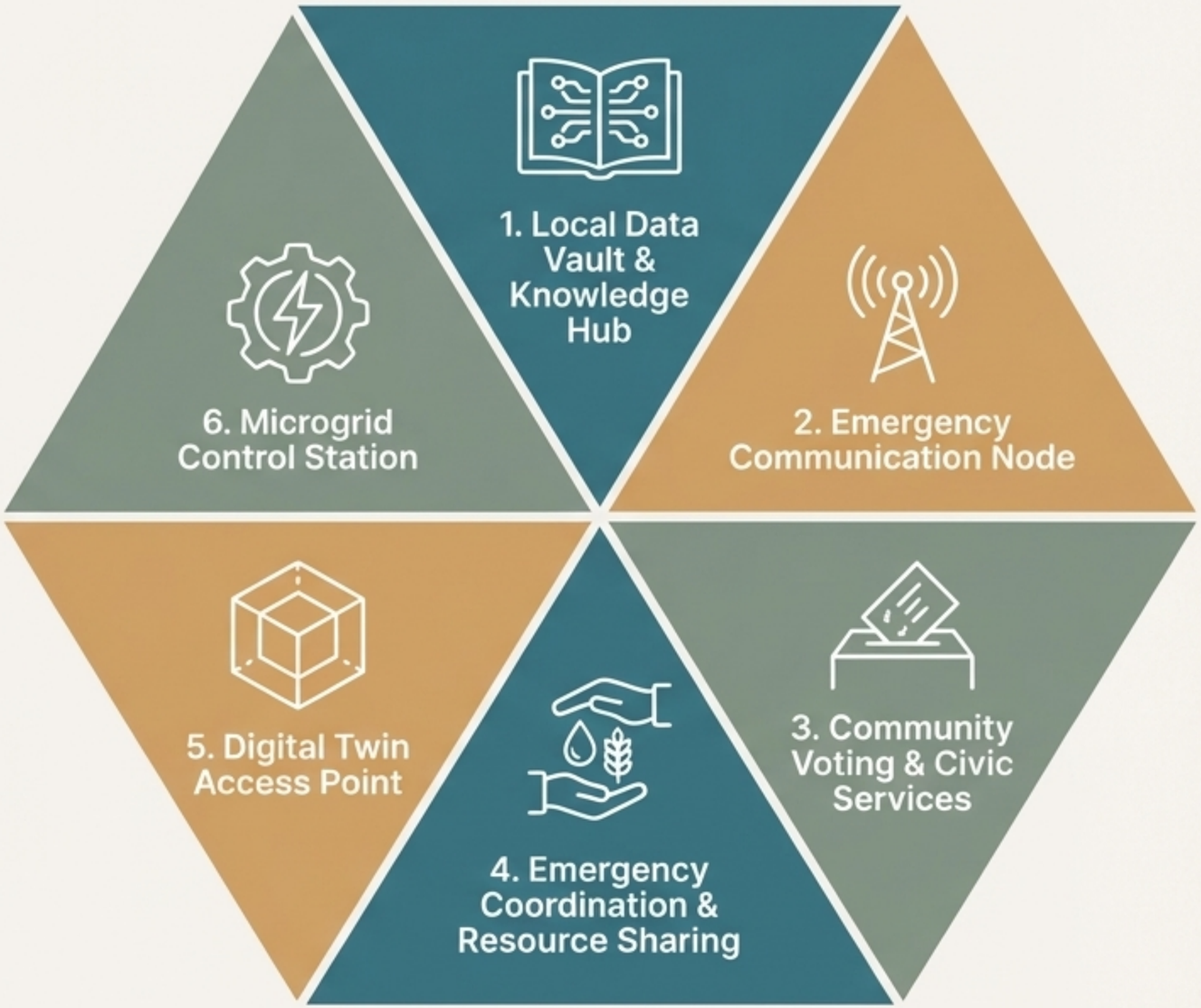


Caption: Kiosks share updates directly, no central server required.

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A Multi-Purpose Hub for Community Resilience

Each kiosk is designed to be a multi-tool, serving day-to-day community needs and providing critical functions during a crisis. This dual-use model ensures the infrastructure is familiar, maintained, and valued year-round.



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Everyday Value: Fostering Knowledge, Culture, and Civic Life



Local Data & Knowledge Hub

Provides offline access to cached maps, emergency manuals, and a digital library. It can host Indigenous knowledge archives, governed by Quandamooka data sovereignty protocols.



Community Voting & Civic Services

Acts as a secure, local “ballot box” for community decisions, surveys, and petitions, running running, running on a transparent community ledger, accessible to all residents.



Digital Twin Access Point

Offers a public interface to “Virtual Minjerribah,” allowing residents to explore planning scenarios, view bushfire or flood models, and engage with the island's digital twin without personal high-end devices.

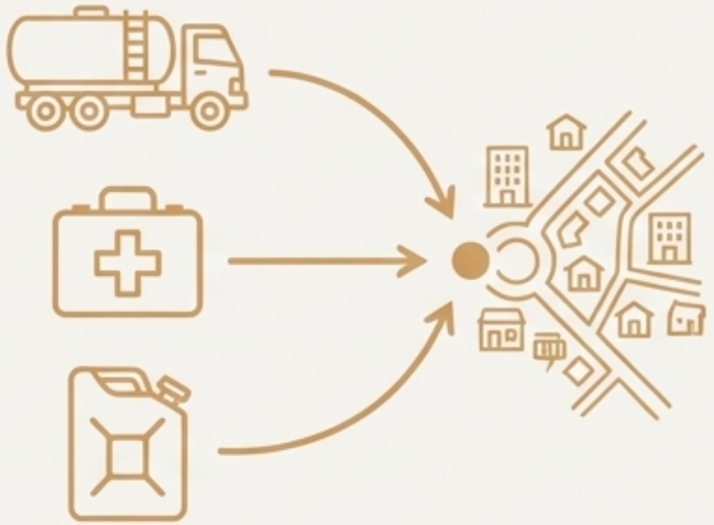
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Crisis Response: A Lifeline for Communication and Coordination



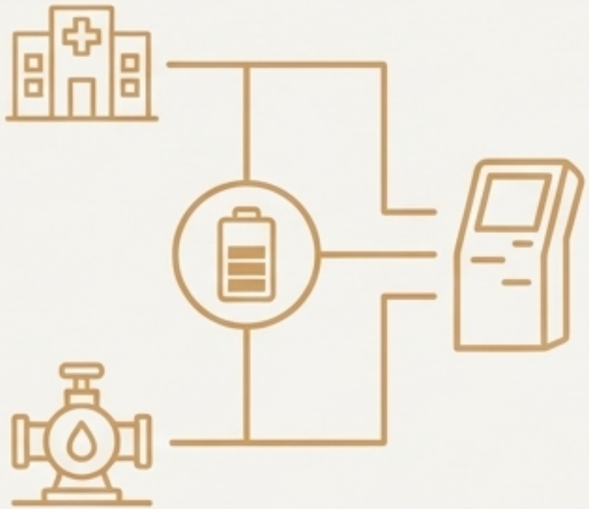
Emergency Communication Node

Broadcasts local alerts, instructions, and community notices. Allows residents to send messages off-island via any available channel (e.g., radio, satellite) and functions as a local message board.



Resource & Coordination Hub

An incident management interface for first responders. Includes a community inventory system or "larder" to log and share available supplies like water, food, and fuel.



Microgrid Control Station

Connects to the community microgrid to display energy status, coordinate load-sharing during outages, and keep critical facilities powered.

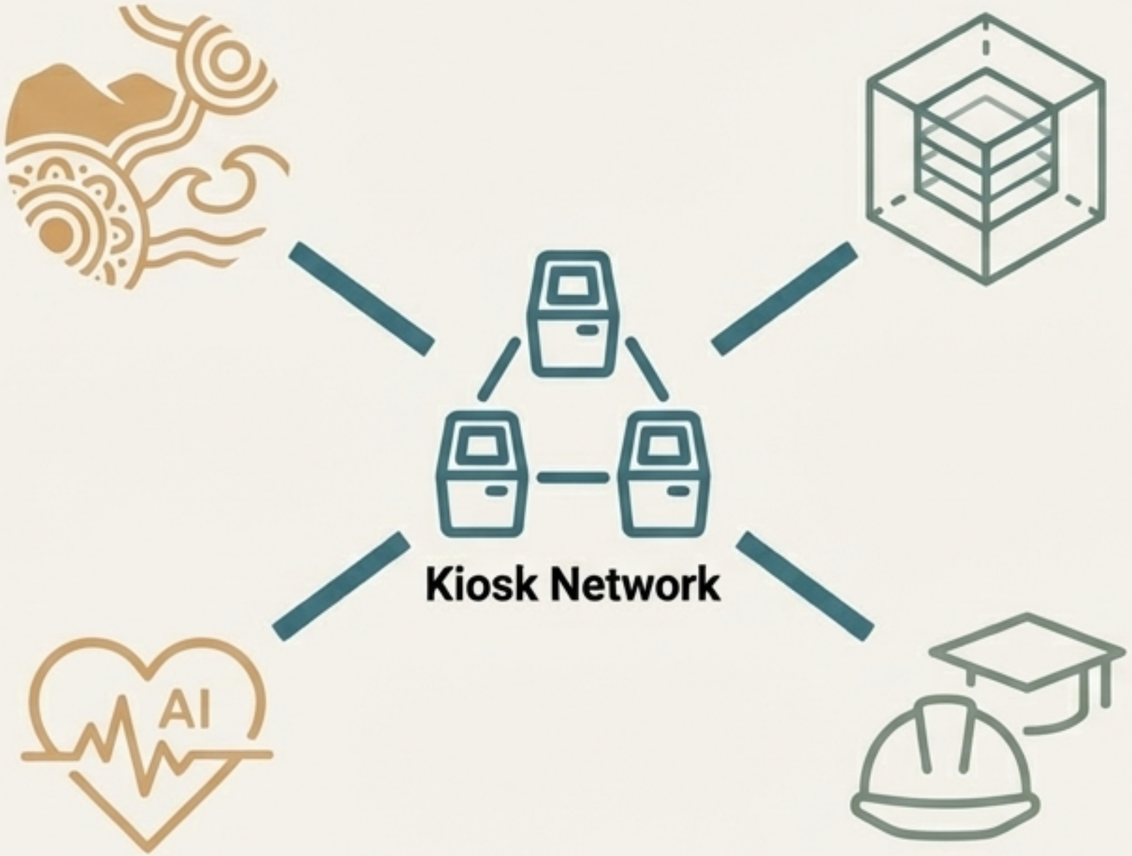
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Integrating with Minjerribah's Innovation Ecosystem

The kiosk plan is designed to connect with and strengthen key local initiatives, creating a cohesive system.

Quandamooka Sovereignty:

Co-designed with QYAC to ensure community ownership and control over data, aligning digital infrastructure with self-determination.



Virtual Minjerribah:

Kiosks act as the physical, offline access points for the island's digital twin, making it accessible to everyone and operational during outages.

Aura of Intelligence:

Provides a platform to host offline 'Aura Capsules' for local health and aged care support, ensuring continuity of care.

SETCo:

Creates skilled local jobs in kiosk deployment and maintenance, with SETCo providing the training and verifiable credentials for the roles.

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A Phased Implementation Strategy

The rollout is structured around flexible milestones, focusing on building capability and community readiness at each stage.



Community Co-Design & Pilot Kiosk

Formalise partnerships (especially with QYAC), host design workshops, and deploy a single functional prototype as a proof of concept.

Core Network & Resilience Features

Expand to 3-5 kiosks covering key settlements, establish the offline mesh network, and conduct disconnection drills to validate performance.

Service Expansion & Integration

Roll out the full suite of functions (Digital Twin, Aura, Microgrid control) and provide community training with SETCo.

Hardening & Scaling Up

Add redundancies, integrate formally into official disaster plans, and run full-scale crisis simulations.

Innovation & Long-Term Sustainability

Use the network as a living lab, secure ongoing funding, and document the model for other communities.

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Potential Funding Pathways for 2025-2026

Strategy: A blended funding approach will be used, combining public grants, research partnerships, and in-kind support. Applications will be timed to align with project milestones.



State Innovation

Advance Queensland programs (Ignite Ideas, Deadly Innovation) for prototyping and Indigenous co-design.



Federal Resilience & Infrastructure

The Disaster Ready Fund (citing the Kowanyama Hub as precedent) and the Thriving Suburbs Program for capital rollout.



Research & Technical Partnerships

CSIRO Kick-Start program for R&D on software and energy systems.



Communications Resilience

Federal programs for mobile network hardening to potentially equip kiosks with satellite links.



Local Support

Redland City Council grants for site preparation and community engagement.

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A Plan for a Self-Reliant and Empowered Minjerribah

The Community Sovereign Kiosk network is a practical investment in the island's future.

- It provides a tangible way to keep Minjerribah connected, coordinated, and powered through crises.
- It fuses the wisdom of self-reliance with robust, decentralised technology.
- Crucially, the plan is built on community ownership, honours Quandamooka sovereignty, and integrates with the island's existing strengths.

This initiative is a tangible step toward the "Joyful, Resilient Abundance" the community is co-creating, turning potential vulnerability into enduring strength.



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